

Upgrading of methanol and ethanol to higher alcohols over NixCe/Mg₂Al-LDO catalysts

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ABSTRACT

Catalytic conversion of methanol and ethanol through the Guerbet reaction is one of the promising ways to produce the value-added higher alcohols. It is the core issue to develop the highly efficient catalyst for satisfying the commercialization of the technology. In this work, NixCe/Mg₂Al-LDO (Layered Double Oxides) catalysts with Ce/Ni molar ratios of 0, 0.15, 0.3 and 0.5 were prepared by co-impregnation method for the upgrading of methanol and ethanol to C₃₊OH. The highest selectivity of C₃₊OH reached 81.0 % with ethanol conversion of 16.0 % over Ni₃₀Ce/Mg₂Al-LDO catalyst at 320 °C, 0.1 Mpa. Combined with the characterization analysis of the catalyst, we found that the introduction of appropriate amount of Ce optimizes the acid-base properties of the NixCe/Mg₂Al-LDO catalyst and increases the Ni⁰ fraction in the total Ni species, which can effectively facilitate the dehydrogenation and hydrogenation steps during the cross-coupling of methanol and ethanol. Both the high ratio of Ni⁰ in Ni species, Ce³⁺/Ce⁴⁺ redox pair and the moderate proportion of strong base/medium-strong acid sites are beneficial to the conversion of methanol and ethanol to C₃₊OH.

1. Introduction

Methanol and ethanol are cheap bulk alcohol chemicals in the world chemical industry. Upgrading these raw materials to obtain high value-added downstream products has gained much attention in both industrial and academic fields. Higher alcohols (C₃₊ carbon-chain alcohols) possess higher energy density and poorer water-solubility compared with methanol and ethanol, which can be used as fuel additive, bulk fine chemicals for plasticizers, surfactants and pharmaceutical intermediates [1,2]. However, it is very difficult to achieve the conversion of methanol and ethanol to higher alcohols, which usually requires harsh reaction conditions (high pressure and high temperature). The Guerbet reaction may be one of the promising ways to upgrade short chain alcohols to higher alcohols under mild condition. Methanol cannot undergo the self-condensation reaction because its molecule lacks β-H. So, with the aid of β-H in ethanol molecule, the cofeeding of methanol and ethanol can proceed cross-coupling process to obtain higher alcohols. Generally, the Guerbet reaction with respect to the cross coupling of methanol and ethanol can be divided into three essential steps [2–4]: (1) methanol and ethanol are dehydrogenated to formaldehyde and acetaldehyde, respectively; (2) formaldehyde and acetaldehyde condense and

dehydrate to produce acrolein or crotonaldehyde; (3) finally hydrogenated to C₃₊ alcohols, such as propanol, butanol, isobutanol and so on. So the catalysts used for such above reaction usually require the synergistic effect of acidic-basic and metallic active sites to accomplish the dehydrogenation, condense and hydrogenation process.

Up to now, numerous homogeneous catalysts [5–9] and heterogeneous catalysts [3,4,10–29] have been reported for the condensation reaction. The former catalysts once have been widely studied because of their higher activity and selectivity in homogeneous liquid phase during the reaction. King et al. prepared various rhenium complexes used to convert methanol and ethanol into isobutanol, which are supported by bidentate and tridentate phosphinoamine ligands. The optimized catalyst gave 35 % isobutanol yield with a 97 % liquid fraction selectivity at 180 °C [5]. Liu et al. realized a 42 % ethanol conversion and 96 % isobutanol selectivity below 200 °C over a non-noble-metal manganese complex catalyst [6]. Although these developed homogeneous catalysts exhibited the higher alcohol selectivity of the catalytic coupling reaction, the difficult separation of the catalysts and products as well as the consumption of liquid alkali phase resulted in the rapid deactivation of the catalysts, unsatisfying the wide industrial application in cross coupling of lower carbon alcohols.

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Alternatively, various heterogeneous catalysts with better stability have been reported in alcohol coupling reaction, which mainly include hydroxyapatite (HAP) [11,13–22], supported catalysts [23–29] (such as catalysts supported on carbon, Al_2O_3 and basic zeolites) and mixed metal oxides [10,12,30–39]. Among them, the mixed metal oxides derived from layered double hydroxides (LDHs) have been widely concerned, due to their large surface area which favors to the dispersion of metals loaded, and the adjusted acid-basic property, exhibiting good catalytic performance to promote the aldol condensation. Stošić et al. [3] prepared MgAlOx catalysts for the upgrading of methanol and ethanol and the best catalyst at Mg/Al molar ratio of 4 gave a selectivity of 16.76 % toward C_3+OH with methanol and ethanol conversion of 22.23 % and 23.53 % at 400 °C (the molar ratio of methanol/ethanol=1/1), respectively. Dehydrogenation is the first step of Guerbet reaction. Introducing transition metals can enhance the dehydrogenation ability to improve the conversion of alcohol. Bravo-Suárez et al. reported Cu11MgAlOx catalyst for the condensation of methanol and ethanol, which showed 65.0 % ethanol conversion with almost 40 % selectivity of aldehyde and only 7 % selectivity of C_3+OH at 250 °C [12]. Cheng et al. also reported CuMgAlOx ($\text{M}^{2+}/\text{M}^{3+}=4$), achieving an ethanol conversion of 43.92 % with 55.52 % C_3+ aldehyde and 33.71 % C_3+OH selectivity [10]. It can be seen that Cu-based mixed metal oxides have excellent dehydrogenation ability and perform higher ethanol conversion and selectivity to aldehydes, but the poor balance between basic and acidic sites leads to low selectivity of C_3+OH . Compared with the copper-based catalysts, the nickel-containing catalysts possess excellent selectivity towards higher alcohols. Pang et al. reported that Ni_4MgAl exhibited a C_{4-8}OH selectivity up to 85 % at 250 °C, 3 Mpa with an ethanol conversion of 18.7 % in ethanol to butanol [40]. According to the characterization results, the metal Ni promotes ethanol dehydrogenation. However, the powerful cleaving ability of C—C bond over Ni active sites leads to the production of CO, CO_2 and CH_4 while promoting ethanol dehydrogenation, reducing the total carbon efficiency of the reaction. Then some researchers tried to add the second element through adjusting the electronic structure of Ni to improve the higher alcohols selectivity of catalysts. Wu et al. achieved 93.8 % higher alcohol selectivity according to restrict the cracking of C—C bond over Ni-based catalyst by adding Sn [37]. Additionally, CeO_2 with abundant basic sites has been used as an active catalyst to promote the aldol condensation. Li et al. found that adding CeO_2 not only favored the aldol condensation but also improved the dispersion of metal due to the strong interaction of metal and CeO_2 species over $\text{M-CeO}_2/\text{AC}$ ($\text{M}=\text{Cu}, \text{Fe}, \text{Co}, \text{Ni}, \text{Pd}$) [28,41]. Furthermore, Wang et al. also reported that the interaction of Ni and CeO_2 impacted the electronic structure of Ni over NiCeO_2/C catalysts which achieved a 89.7 % selectivity towards C_{4-8}OH [42].

Combining the virtue of MgAl layered double hydroxides and NiCe composites, a series of $\text{NiCe}/\text{Mg}_2\text{Al-LDO}$ catalysts were synthesized by co-impregnation method in the present work. The surface properties, geometric structure and electronic structure of $\text{NiCe}/\text{Mg}_2\text{Al-LDO}$ were characterized using BET, XRD, TEM, $\text{NH}_3/\text{CO}_2\text{-TPD}$ and XPS. In-situ DRIFTS was employed to elucidate the mechanism between the surface properties and catalytic performance. According to the characterization results, $\text{Mg}_2\text{Al-LDO}$ provided a high surface area for metal dispersion of and numerous strong basic and acidic sites. Adding Ce can not only adjust the electronic structure of Ni, but also regulate the acid-

base distribution, synergistically accelerating the dehydrogenation and C—C coupling reaction, thereby achieving high-efficiency conversion to C_3+OH under mild reaction conditions.

2. Experimental

2.1. Catalyst preparation

2.1.1. $\text{Mg}_2\text{Al-LDHs}/\text{LDO}$

Mg_2Al -Layered Double Hydroxide (LDHs) were produced using the co-precipitation method. Solution A: The molar concentration of $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (5.13 g) and $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (3.75 g) is 0.3 mol/L; Solution B: The molar concentration of NaOH (2.4 g) and Na_2CO_3 (1.064 g) is 0.7 mol/L. Both of solutions A and B were added to a three-necked flask at 30 °C. Following adjusting the pH of the resulting suspension to 10 by NaOH solution (2 mol/L). $\text{Mg}_2\text{Al-LDHs}$ can be obtained after washing ($\text{pH}=7$) and drying overnight at 60 °C. Subsequently, these materials were calcined at 500 °C for 2 h (2 °C/min), producing $\text{Mg}_2\text{Al-Layered Double Oxides (LDO)}$.

2.1.2. $\text{NiCe}/\text{Mg}_2\text{Al-LDO}$

$\text{NiCe}/\text{Mg}_2\text{Al-LDHs}$ were synthesized by the co-impregnation method. First, 0.1982 g of $\text{Ni}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ and the required amount of $\text{Ce}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ were dissolved in ethanol (10 mL). Then, $\text{Mg}_2\text{Al-LDHs}$ (4.0 g) were added into the above solution and agitated at 30 °C for 4 h, followed by drying at 60 °C overnight. The resulting powder was denoted as $\text{NiCe}/\text{Mg}_2\text{Al-LDHs}$, and x represents the molar ratios of Ce/Ni, corresponding to theoretical Ce loadings of 15, 30 and 50 % based on Ni, respectively. The theoretical loading of Ni was 1 wt % based on $\text{Mg}_2\text{Al-LDHs}$. Subsequently, $\text{NiCe}/\text{Mg}_2\text{Al-LDO}$ was obtained by calcining the $\text{NiCe}/\text{Mg}_2\text{Al-LDHs}$ at 500 °C for two hours (2 °C/min).

2.2. Catalyst characterization

BET, XRD, TEM, $\text{NH}_3/\text{CO}_2\text{-TPD}$, XPS and In-situ DRIFTS were used to characterized the catalysts. Detailed process of characterization was shown in the Supplementary Information.

2.3. Catalytic reaction

All experiments were carried out using 0.80 g of catalyst (40–60 mesh) diluted with quartz sand to 3 mL in a fixed-bed reactor (the inner diameter of reactor tube is 8 mm) under 0.1Mpa. Prior to reaction, reducing the catalyst at 500 °C for 1 h with 20 vol % H_2 in N_2 . Following the temperature dropped to reaction temperature with N_2 (60 mL/min), the liquid of mixed methanol and ethanol (methanol/ethanol molar ratio=3/1, LHSV=3.76 $\text{mL} \cdot \text{gcat}^{-1} \cdot \text{h}^{-1}$) was pumped into the evaporator and then preheated to 150 °C, finally introduced into the reactor by N_2 . The gaseous products (H_2 , CO, gas hydrocarbons and CO_2) were analyzed by an HaiXin gas chromatograph (GC) with a thermal conductivity detector (TCD) and a flame ionization detector (FID). An Al_2O_3 column with FID (carrier gas: Ar) was used to analyze the lower hydrocarbons (C1–C5). H_2 , CO, CO_2 , and CH_4 were analyzed using a Carbosieve-packed column with TCD. The liquid products (alcohols, ethers, esters) were examined by an Agilent 7890A GC (column:DB-FFAP) with FID and TCD. The GC oven temperature program was set as

follows: an initial hold at 70 °C for 1 min, followed by a ramp to 180 °C at a rate of 5 °C/min, and a final hold at 180 °C for 5 min. The column flow rate was maintained at 25 mL/min (N₂). The detector temperatures were set at 250 °C. The carbon balance of this reaction over Ni₃₀Ce/Mg₂Al-LDO catalysts was better than 95 %. The conversion and product selectivity were calculated as follows:

$$\text{Methanol conversion}(\%) = \frac{100 \times (\text{C moles in reacted methanol})}{(\text{C moles in products from methanol} + \text{C moles in unreacted methanol})} \quad (1)$$

$$\text{Ethanol conversion}(\%) = \frac{100 \times (\text{C moles in reacted ethanol})}{\text{C moles in products from ethanol} + \text{C moles in unreacted ethanol}} \quad (2)$$

$$\text{Product } i \text{ selectivity } (\%) = \frac{100 \times (\text{C moles in product } i)}{\text{C moles in all products}} \quad (3)$$

$$\text{The ratio of cross-coupling} = \frac{\text{Selectivities of (propanol + isobutanol + pentanol)}}{\text{Selectivity of butanol}} \quad (4)$$

3. Results and discussion

3.1. Catalytic conversion of methanol and ethanol

The Ni₃₀Ce/Mg₂Al-LDO catalysts with different Ce contents were applied in the cross-coupling of methanol and ethanol at 320 °C, 0.1 MPa (Table 1). It can be seen that the ethanol conversion of Mg₂Al-LDO is only 4.1 %. The products mainly contain C₃₊OH, ether and gas, with selectivity is 40.4 %, 36.5 %, and 6.5 %, respectively. The selectivity of ether is almost similar to that of C₃₊OH. The ratio of cross-coupling of methanol and ethanol is only 1.1, indicating the reactivity of methanol and ethanol is comparable. However, the ethanol conversion shows an increase from 4.1 % to 9.4 % and the product distribution also changes, the selectivity of ether and gas decrease significantly, while the selectivity of C₃₊OH dramatically increases to 80.2 % over Ni/Mg₂Al-LDO. At the same time, the secondary condensation products, such as isobutanol and C₅OH, are formed and the ratio of cross-coupling of methanol and ethanol is improved to 1.6, indicating the enhanced ability of dehydrogenation and condensation for alcohol reactants. As for Ni₃₀Ce/Mg₂Al-LDO sample, the ratio of cross-coupling of methanol and ethanol further increases to 2.1, demonstrating that the preferential tendency to participate in the reaction for methanol is enhanced with higher Ce content.

Meanwhile, the ethanol conversion increases firstly and then decreases, showing a volcanic trend. Over the optimal Ni₃₀Ce/Mg₂Al-LDO catalyst, 16.0 % of ethanol conversion and 81.0 % of C₃₊OH selectivity

are achieved, respectively. In general, Ni₃₀Ce/Mg₂Al-LDO exhibits higher selectivity toward secondary condensation products (isobutanol and pentanol) compared to Ni/Mg₂Al-LDO, indicating that the appropriate Ce enhances the condensation ability of Ni₃₀Ce/Mg₂Al-LDO.

The effect of LHSV(7.50, 3.76, 2.25, 1.5 mL·gcat⁻¹·h⁻¹) on the condensation products also was investigated, and the result are shown in

Figure.S1A. We observed that decreasing the LHSV led to higher conversions of both methanol and ethanol, while the highest C₃₊OH selectivity was achieved at 3.76 mL·gcat⁻¹·h⁻¹. The Methanol/Ethanol ratio (M/E) plays a crucial role in determining the cross-condensation selectivity and consequently influences the product distribution. Therefore, we systematically investigated the effect of varying M/E ratios (6:1, 3:1, 1:1, 1:3) on the reaction performance, as shown in Figure. S1B. With increasing M/E ratio, methanol conversion decreases while ethanol conversion rises, attributed to the reduced ethanol concentration in the feed stream. The C₃₊OH selectivity exhibits a volcano-type dependence, reaching its maximum at a M/E ratio of 3:1. The selectivities toward propanol, isopropanol, and pentanol increased, accompanied by reduced butanol formation. The observed enhancement in cross-condensation ratio clearly demonstrates a preference for methanol-ethanol cross-condensation over ethanol self-condensation pathways. However, further increasing M/E ratio, the amount of ethanol involved in cross-coupling and self-condensation reduces, leading to the decreasing space-time yield of C₃₊OH. Based on the above results, we keep the M/E ratio of 3:1 and LHSV of 3.76 as the optimal condition for this catalytic system.

The product distribution and conversion are sensitive to temperature. Therefore, the influence of temperature on catalytic performance over Ni₃₀Ce/Mg₂Al-LDO was studied. As shown in Fig. 1A, with the temperature varying from 320 °C to 380 °C, the conversion (methanol, ethanol) shows a linear increase and the conversion of ethanol increases from 16.0 % to 34.3 %, but the selectivity for C₃₊ alcohols decreases to 49.6 % due to the generation of gaseous products, such as CO, CO₂ and olefins. According to the literature, dehydrogenation is an endothermic process. Increasing temperature thermodynamically favors the

Table 1
Reactivity of methanol and ethanol over Mg₂Al-LDO and Ni₃₀CeMg₂Al-LDO catalysts.

Catalyst	Conversion(%)		Selectivity(%)								C ₃₊ OH	The ratio of cross-coupling
	MeOH	EtOH	AcH	C ₃ OH	n-C ₄ OH	i-C ₄ OH	C ₅ OH	Ether	Gas	Others		
Mg ₂ Al-LDO	1.2	4.1	6.4	20.7	19.6	-	-	36.5	6.5	10.3	40.4	1.1
Ni/Mg ₂ Al-LDO	1.8	9.4	11.2	35.7	30.4	8.2	5.9	4.8	3.8	-	80.2	1.6
Ni ₁₅ Ce/Mg ₂ Al-LDO	1.5	9.0	8.9	33.8	27.7	7.7	5.0	6.6	1.5	8.8	74.2	1.7
Ni ₃₀ Ce/Mg ₂ Al-LDO	2.0	16.0	16.2	36.9	27.0	10.1	7.0	-	2.8	-	81.0	2
Ni ₅₀ Ce/Mg ₂ Al-LDO	2.1	13.7	10.9	36.1	25.2	9.3	6.0	4.2	1.4	7.0	76.6	2.1

Reaction conditions: 800 mg catalyst, T = 320 °C, P = 0.1 MPa, flowing rate = 60 mL/min, LHSV = 3.76 mL·gcat⁻¹·h⁻¹, Methanol/Ethanol molar ratio= 3/1. (AcH = acetaldehyde; n- = normal; i- = isomeric; OH = alcohol; Gas includes CO, CO₂ and C_xH_y; Others include pentanal, 2-methyl-butanol, butyl acetate, ethyl butyrate, propyl acetate, etc.).

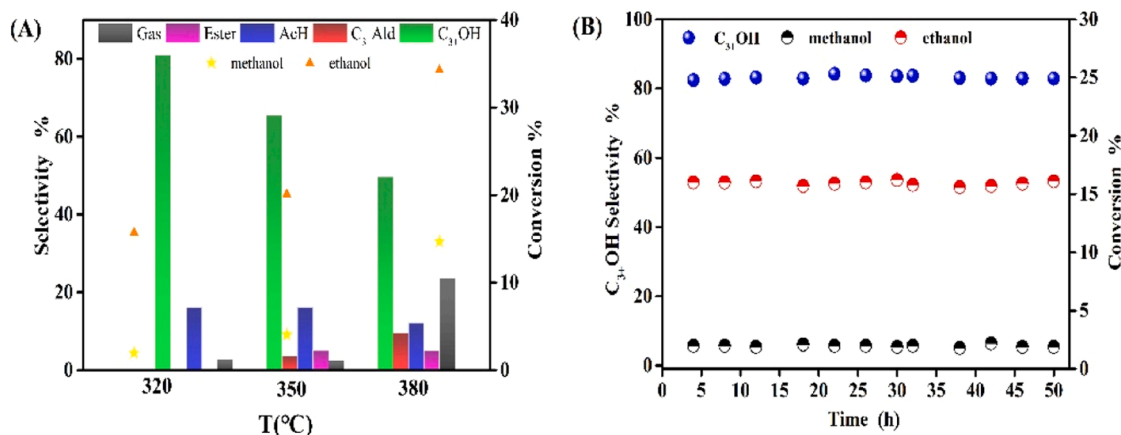


Fig. 1. Methanol and ethanol conversions over the Ni30Ce/Mg₂Al-LDO catalyst at various temperatures (A). Stability of the Ni30Ce/Mg₂Al-LDO for methanol and ethanol to C₃₊ alcohols at 320 °C (B).

Reaction conditions: 800 mg catalyst, $P = 0.1$ MPa, flowing rate = 60 mL/min, LHSV = 3.76 mL·gcat⁻¹·h⁻¹, Methanol/Ethanol molar ratio = 3/1.

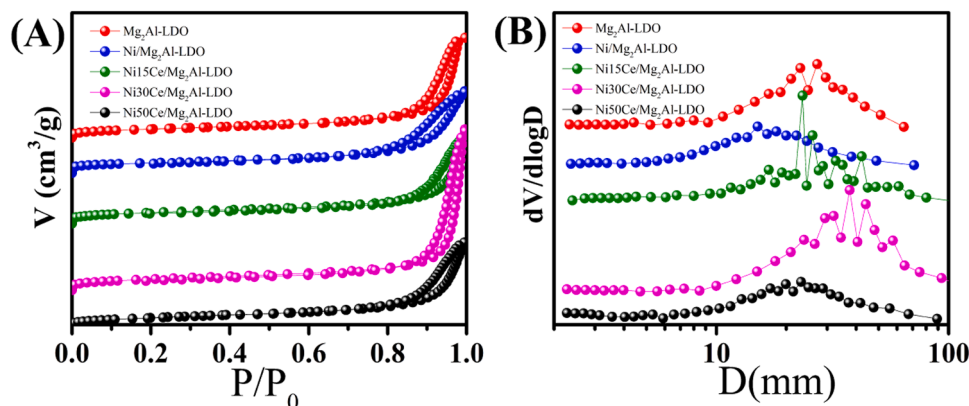


Fig. 2. N₂ adsorption-desorption isotherm (A) and pore size distribution curves (B) of the Mg₂Al-LDO and Ni/Ce/Mg₂Al-LDO catalysts.

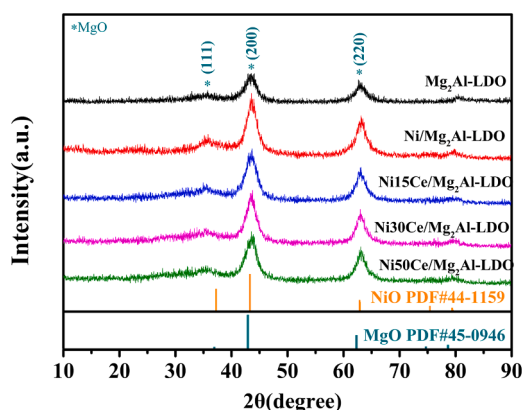


Fig. 3. XRD patterns of the Mg₂Al-LDO and Ni/Ce/Mg₂Al-LDO catalysts.

dehydrogenation, pyrolysis and dehydration [43]. In addition, the stability of Ni30Ce/Mg₂Al-LDO catalyst have been tested, under the reaction conditions: $T = 320$ °C, $P = 0.1$ MPa, flowing rate = 60 mL/min, LHSV = 3.76 mL·gcat⁻¹·h⁻¹, and a M/E molar ratio of 3/1. As shown in Fig. 1B, the almost constant methanol and ethanol conversion (2.0 % and 16.0 %) with C₃₊OH selectivity of 82 % can be observed for Ni30Ce/Mg₂Al-LDO catalyst during the 50 h reaction time, implying the good stability of the sample.

3.2. Catalyst characterization

All samples present Type IV with H2-shaped hysteresis loops and a wide pore size distribution (Fig. 2), suggesting that a mesoporous structure presents in all catalysts and that the structures of these catalysts are not be changed by the addition of Ni and Ce. The specific surface area (S_{BET}) of Mg₂Al-LDO is 210 m²/g (Table S1), which gradually increases to 233 m²/g with the introduction of Ni. However, for Ni/Ce/Mg₂Al-LDO, as the content of Ce increases from 0 to 50 %, the S_{BET} initially increases and then decreases, reaching a maximum of 271 m²/g for Ni30Ce/Mg₂Al-LDO, which may be due to Ni and Ce adsorb on the surface of Mg₂Al-LDH during the synthesis process, and the S_{BET} increases because of the decomposition of nitrate during calcination. In contrast, excessive Ce loading slightly reduces the S_{BET} of Ni/Ce/Mg₂Al-LDO catalysts. The average pore diameter (D_{pore}) and pore volume (V_p) of the Ni/Ce/Mg₂Al-LDO catalyst except for Ni30Ce/Mg₂Al-LDO decrease from 19.8 nm and 1.04 cm³/g to 14.4 nm and 0.89 cm³/g compared to Mg₂Al-LDO (Table S1), because of the deposition of the Ni and Ce species inside the pores of Mg₂Al-LDO.

All catalysts exhibit diffraction peaks at 36.9°, 42.9°, and 62.3°, which can be attributed to the (111), (200) and (220) crystallographic planes of MgO (PDF#45-0946), respectively (Fig. 3). After loading with Ni and Ce, the diffraction peaks of MgO become more intense, demonstrating the strong interaction between Ni and Mg. In addition, no characteristic crystal planes of Al₂O₃, NiO, and CeO₂ are detected, suggesting that Al₂O₃, NiO, and CeO₂ are highly dispersed on catalysts or exist in amorphous forms.

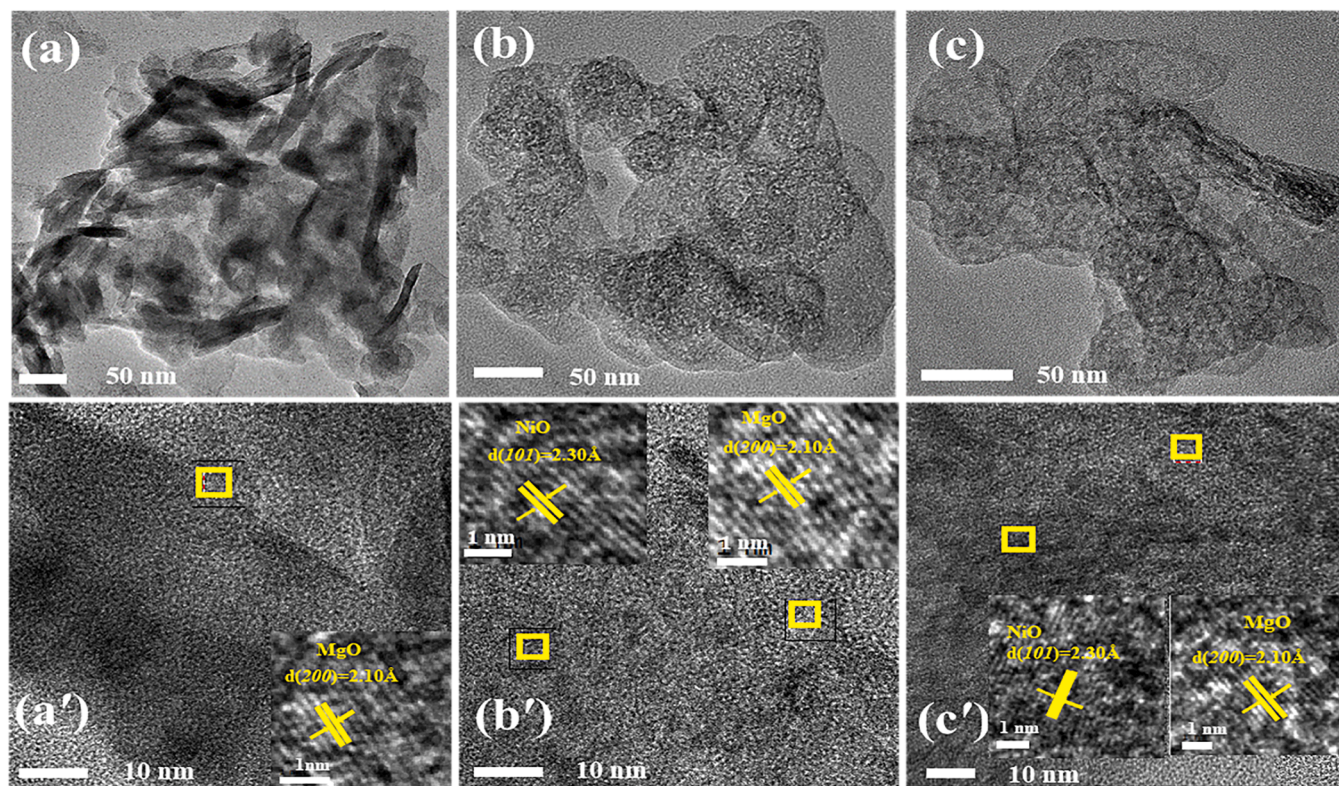


Fig. 4. TEM images of (a-a') $\text{Mg}_2\text{Al-LDO}$, (b-b') $\text{Ni/Mg}_2\text{Al-LDO}$, and (c-c') $\text{Ni}_{30}\text{Ce/Mg}_2\text{Al-LDO}$ catalysts.

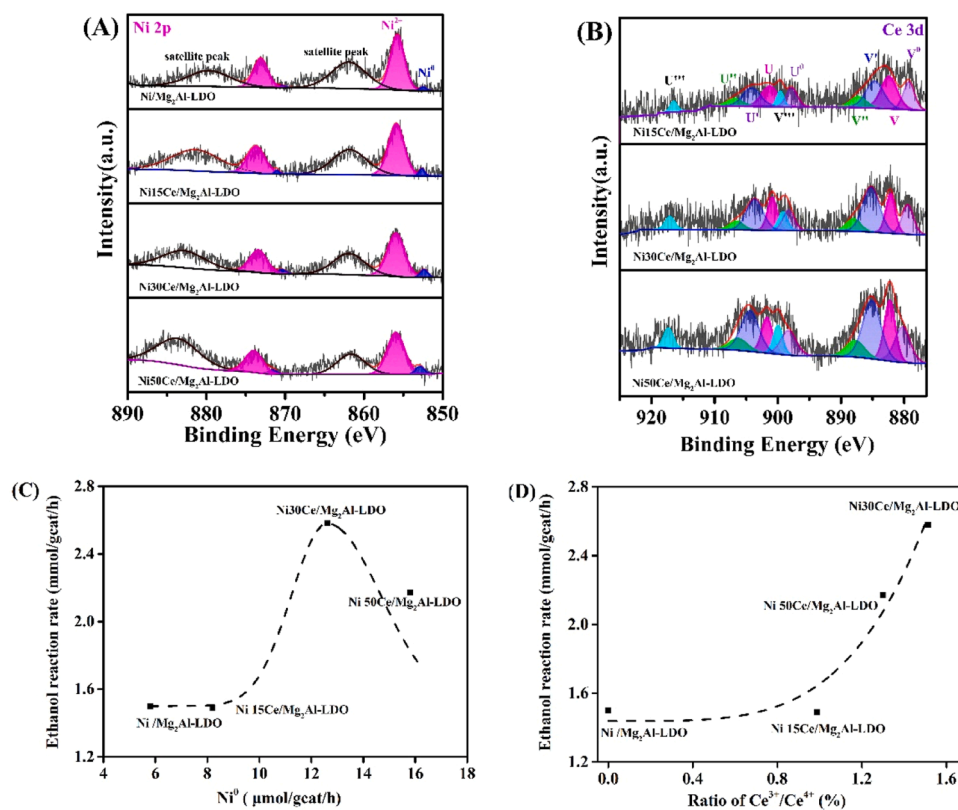


Fig. 5. (A) XPS spectra of Ni 2p and (B) Ce 3d for the $\text{Ni}_x\text{Ce/Mg}_2\text{Al-LDO}$ catalysts (C) Correlation between the reaction rate of ethanol with the amount of Ni^0 and (D) the ratio of $\text{Ce}^{3+}/\text{Ce}^{4+}$. Reaction conditions: $T = 320^\circ\text{C}$, $P = 0.1\text{ MPa}$, flowing rate = 60 mL/min , LHSV = $3.76\text{ mL}\cdot\text{gcat}^{-1}\cdot\text{h}^{-1}$, Methanol/Ethanol molar ratio = $3/1$.

TEM images of $\text{Mg}_2\text{Al-LDO}$, $\text{Ni/Mg}_2\text{Al-LDO}$, and $\text{Ni}_{30}\text{Ce/Mg}_2\text{Al-LDO}$ catalysts are shown in Fig. 4. From Fig. 4a, it is clear that $\text{Mg}_2\text{Al-LDO}$ retains the layered structure of the hydrotalcite precursor. In the large-scale image, the lattice distance of 0.210 nm can be seen, which is related to the MgO (200) plane (Figure 4a'). There are no lattice diffraction stripes of Al_2O_3 can be observed, indicating high dispersion of Al_2O_3 , which agrees with the XRD results [43]. No obvious change in the layer structure can be seen after nickel deposition (Fig. 4b). In Figure 4b', the lattice spacing of 0.210 and 0.230 nm are assigned to the plane of MgO (200) and NiO (101), respectively. For $\text{Ni}_{30}\text{Ce/Mg}_2\text{Al-LDO}$ catalysts, the morphology is similar to that of $\text{Ni/Mg}_2\text{Al-LDO}$. In Fig. 4c', only the MgO (200) plane and the NiO (101) plane are observed, while CeO_2 particles are not be clearly visible, demonstrating the excellent dispersion of CeO_2 , in agreement with the XRD characterization results.

To study the chemical states of Ni and Ce, XPS was conducted on the $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ catalysts. The XPS spectra of Ni 2p and Ce 3d are shown in Fig. 5. Simultaneously, the valence states of the elements were quantified by normalizing the area of the characteristic peaks and listed in Table S2. Ni 2p spectrum is mainly composed of Ni 2p_{3/2}, Ni 2p_{1/2} signal peaks, and their corresponding satellite peaks (861.90 eV, 879.66 eV). The peaks located at 852.49 and 870.44 eV are corresponding to the Ni 2p_{3/2} and Ni 2p_{1/2} spin-orbit splitting peaks of metallic Ni^0 . The peaks at 855.75 eV and 873.10 eV belong to Ni^{2+} of Ni 2p_{3/2} and Ni 2p_{1/2}, respectively. From the XPS spectrum, we know that nickel species on the surface mainly exist in an oxidized state, and the metal state accounts for a relatively small proportion (Table S2). The ratio of Ni^0 on $\text{Ni/Mg}_2\text{Al-LDO}$ is only 3.4 %. It is because the strong interaction between Ni and Al or Mg, as confirmed by the XRD results, making it difficult to reduce the oxidized nickel. With the addition of Ce, the content of Ni^0 gradually increases, reaching the highest proportion of 9.5 % over $\text{Ni}_{50}\text{Ce/Mg}_2\text{Al-LDO}$. This phenomenon can be explained by that Ce adjusting the electronic environment around Ni and weakening the interaction between Ni and Mg or Al, which is conducive to the reduction of Ni^{2+} [44]. Besides, the satellite peak located at 879.66 eV shifts to a higher binding energy (883.60 eV) with the increase in Ce loading, which is correlated to the similar binding energy between the Ni 2p_{1/2} satellite peak and Ce 3d_{5/2} [45,46].

Fig. 5B is the XPS spectrum of Ce 3d. The Ce 3d spectrum is decomposed into ten peaks, labeled as V^0 (879.25 eV), V (882.39 eV), V' (884.66 eV), V'' (887.24 eV), V''' (899.58 eV), U^0 (897.78 eV), U (901.28 eV), U' (904.15 eV), U'' (906.8 eV), U''' (916.46 eV), respectively. V^0 , V' , and U^0 , U' correspond to the 3d_{5/2} and 3d_{3/2} spin-orbit splitting peaks of

Ce^{3+} species, while V'' , V''' , and U , U' , U''' can be ascribed to the 3d_{5/2} and 3d_{3/2} spin-orbit splitting peaks of the Ce^{4+} species, which is in accordance with the literature [29,42]. This indicates that both Ce^{3+} and Ce^{4+} exist in the $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ catalysts. The minimum proportion of Ce^{3+} in $\text{Ni}_{15}\text{Ce/Mg}_2\text{Al-LDO}$ is 49.7 %. As the Ce loading increases to 30 %, the ratio of Ce^{3+} increases to 60.2 %. This can be explained by the interaction between Ni and Ce, the nearby Ce^{4+} is easily reduced to Ce^{3+} when H_2 dissociates on Ni^0 with the ratio of Ce^{3+} increases, because of the hydrogen spillover effect [44]. As the content of Ce increases to 50 %, the relative amount of Ni decreases, and the degree of reduction declines, leading to a decrease in the proportion of Ce^{3+} to 56.5 %.

Dehydrogenation serves as the first step in the Guerbet reaction. According to the literature, a high ratio of $\text{Ce}^{3+}/\text{Ce}^{4+}$ redox pairs constructs more defects and is beneficial for electron transfer [42]. At the same time, it is known that Ni^0 favors the dehydrogenation of methanol and ethanol [37,47]. Our investigation establishes a correlation between the ethanol reaction rate and two key parameters: Ni^0 content and $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio (Fig. 5C and D). Fig. 5C demonstrates a characteristic volcano-shaped dependence of the ethanol reaction rate on Ni^0 content, clearly indicating the involvement of additional limiting factors beyond Ni^0 . However, Fig. 5D reveals a strong positive correlation between ethanol reaction rate and the $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio, with higher ratios consistently promoting ethanol conversion. High $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio and Ni^0 are beneficial for the dehydrogenation of methanol and ethanol. Notably, the $\text{Ce}^{3+}/\text{Ce}^{4+}$ ratio exerts a more pronounced catalytic effect than Ni^0 alone. This mechanistic insight explains the superior performance of $\text{Ni}_{30}\text{Ce/Mg}_2\text{Al-LDO}$ over its $\text{Ni}_{50}\text{Ce/Mg}_2\text{Al-LDO}$, despite the latter's higher Ni^0 loading.

The acid-base sites play a key role in the aldol condensation of methanol and ethanol, so the CO_2/NH_3 -TPD of $\text{Mg}_2\text{Al-LDO}$ and $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ were characterized (Fig. 6). Fig. 6A is the NH_3 -TPD profiles of $\text{Mg}_2\text{Al-LDO}$ and $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ catalysts. It can be seen that the NH_3 -TPD profile of all $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ catalysts exhibit a broad NH_3 desorption between 50 and 400 °C, which can be divided into three Gaussian peaks: weak acid sites (A_w , -OH) at 120 °C, medium strong acid sites (A_{ms} , $\text{Al}^{3+}\text{-O-M}^{2+}$, $\text{M}=\text{Mg, Ni}$) at 180 °C, and strong acid sites (A_s , Al^{3+}) at >310 °C [35,39,48], respectively. While only weak and strong acid sites on $\text{Mg}_2\text{Al-LDO}$ with 3.709 mmol/g of the amount of total acid sites are detected, and the ratio of strong acidic sites accounts for up to 92.2 % (Table 2). According to the literature, strong acidic sites are beneficial for intermolecular and intramolecular dehydration to produce

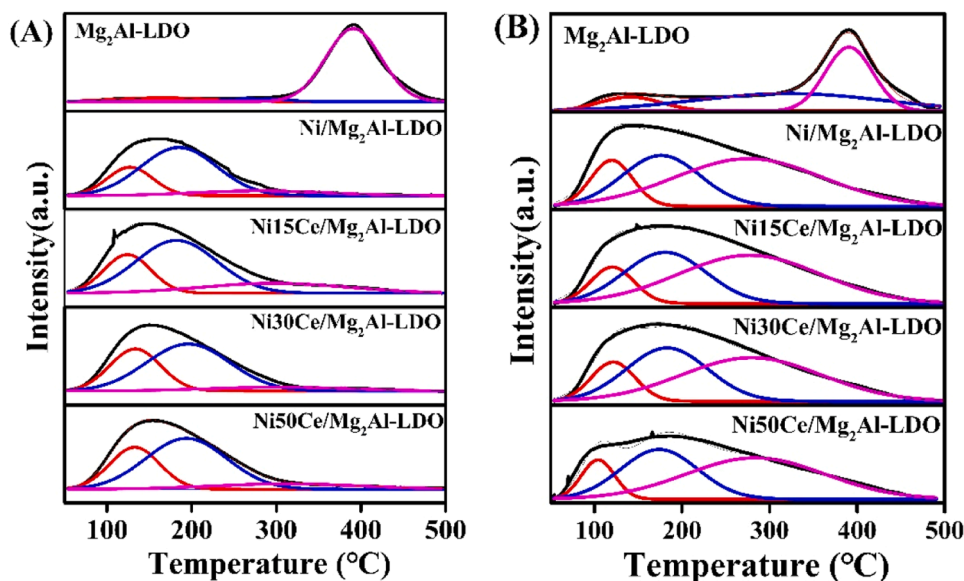


Fig. 6. (A) NH_3 -TPD and (B) CO_2 -TPD patterns of the $\text{Mg}_2\text{Al-LDO}$ and $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ catalysts with different Ce loadings.

Table 2
Acid-base properties of the Mg₂Al-LDO and Ni/Ce/Mg₂Al-LDO catalysts.

Sample	Acid sites /mmol·g ⁻¹	Acid sites / mmol·g ⁻¹			Basic sites /mmol·g ⁻¹	Basic sites/ mmol·g ⁻¹		
		A _w	A _{ms}	A _s		B _w	B _{ms}	B _s
Mg ₂ Al-LDO	3.709	0.288 (7.8)	–	3.421 (92.2)	2.547	0.311 (12.2)	–	2.236 (87.8)
Ni/Mg ₂ Al-LDO	0.510	0.114 (22.4)	0.332 (65.1)	0.064 (12.5)	0.851	0.130 (14.9)	0.267 (30.6)	0.476 (54.5)
Ni15Ce/Mg ₂ Al-LDO	0.404	0.097 (24.0)	0.23 (56.9)	0.077 (19.1)	0.732	0.091 (12.5)	0.243 (33.2)	0.398 (54.3)
Ni30Ce/Mg ₂ Al-LDO	0.398	0.129 (32.4)	0.236 (59.4)	0.033 (8.27)	0.700	0.099 (14.1)	0.250 (35.7)	0.351 (50.1)
Ni50Ce/Mg ₂ Al-LDO	0.400	0.120 (29.8)	0.237 (59.2)	0.044 (11.0)	0.642	0.077 (12.0)	0.226 (35.2)	0.339 (52.9)

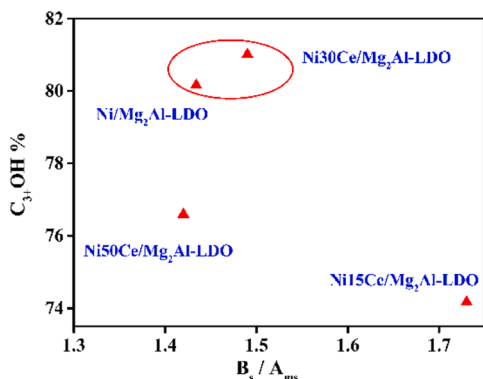


Fig. 7. Selectivity of C₃₊OH as a function of the ratio of strong base to medium-strong acid sites.

ether and olefin [49], respectively, which is consistent with the selectivity of ether in this work (Table 1, entry 1). Quantitatively speaking, after loading Ni, the number of total acid sites and strong acid sites dramatically decreases from 3.709 to 0.51 mmol/g, and 3.421 to 0.064 mmol/g, respectively. However, the proportion of A_{ms} increases to 65.1 %, which can be deduced that Ni mainly covers on the strong acidic sites Al³⁺, leading to the formation of medium-strong acidic sites (Al³⁺-O-Ni²⁺) [40,50]. For Ni/Ce/Mg₂Al-LDO, the number of total acid sites and medium-strong acid sites slightly decreases compared to Ni/Mg₂Al-LDO.

From Fig. 6B, it can be seen that the CO₂-TPD profiles of all catalysts exhibit a broad CO₂ desorption between 50 and 500 °C, which can be fitted into three peaks: weak basic sites (B_w, surface basic -OH groups) at 120 °C, medium-strong basic sites (B_{ms}, M⁺-O²⁻, M=Mg, Ni) at 180 °C, and strong basic sites (B_s, O²⁻) at >280 °C [31], respectively. However, only weak and strong basic sites are observed on Mg₂Al-LDO, with a high ratio of strong basic sites, 87.8 % (Table 2). Loading Ni, the number of total basic sites decreases significantly to 0.851 mmol/g, which may be due to Ni covering some basic sites, leading to a reduction in the total number of basic sites during preparation by the co-impregnation method [51]. At the same time, Ni interacts with the surface strong basic site O²⁻ to form a medium-strong basic site Ni²⁺-O²⁻, resulting in an increased proportion of B_{ms} and a decreased proportion of B_s. For Ni/Ce/Mg₂Al-LDO, the number of total basic sites decreases slightly with the Ce addition. However, the proportion of B_{ms} increases, showing a volcanic trend. The highest ratio of B_{ms} is 35.7 % over Ni30Ce/Mg₂Al-LDO.

The aldol condensation is a key step in the C—C coupling to produce C₃₊OH. The condensation product is strongly influenced by the number and distribution of acid-base sites. According to the literature, Lewis acid sites can adsorb and stabilize ethoxide intermediates [50] and strong basic sites are beneficial for the condensation of aldehydes [13,42]. A suitable ratio of strong basic/medium-strong acid pairs (B_s/A_{ms}) are conducive to the condensation reaction [40]. Therefore, we correlated

the ratio of B_s/A_{ms} with the selectivity of C₃₊OH. The results are shown in Fig. 7. Ni30Ce/Mg₂Al-LDO possesses a moderate ratio of B_s/A_{ms} with higher C₃₊OH selectivity, suggesting that there is a synergistic catalytic effect between the basic and acid sites in the condensation process. At a high B_s/A_{ms} ratio, a low concentration of adsorbed aldehyde results in a low amount of condensation products. However, a low ratio of B_s/A_{ms} also leads to lower selectivity of C₃₊OH due to weak coupling ability. We have also conducted additional calculations of the product formation rates and normalized these values with respect to both strong basic sites and medium-strong acid sites (Table S3). It clearly demonstrates that the incorporation of Ce significantly enhances the normalized formation rate of C₃₊OH products. The appropriate addition of Ce can not only optimize the acid-base distribution but also promote the balance of acid-base sites, which is necessary for the aldol reaction, thereby increasing the selectivity and yield of C₃₊OH [39].

Based on the above characterization results, it can be seen that the introduction of Ni on Mg₂Al-LDO optimizes the acid-base sites to improve the C—C coupling ability and the selectivity of C₃₊OH. At the same time, the presence of metallic Ni accelerates the dehydrogenation process and thereby improves the ethanol conversion. Introducing Ce to Ni/Mg₂Al-LDO increases the ratio of Ni⁰ by adjusting the electronic environment around Ni, and the Ce³⁺/Ce⁴⁺ redox pair benefits electron transfer. Both of these factors favor the dehydrogenation of methanol and ethanol. Furthermore, the presence of Ce regulates the acid-base sites of the catalysts to help to improve the selectivity and yield of C₃₊OH.

3.3. Mechanistic investigation

To reveal the role of Ni and Ce in the coupling of methanol and ethanol, in situ DRIFTS spectra were recorded (Fig. 8). All catalysts show absorption bands at 2982, 2936, 2866, 2813, 1277, 1118, 1058 cm⁻¹ at 50 °C. According to the literature, the bands at 2982, 2866, 2936, 2813 cm⁻¹ correspond to the stretching modes of ν_{as}(CH₃), ν_s(CH₃), ν_{as}(CH₂) and ν_s(CH₂) of aliphatic alcohols, respectively [35]. The band at 1277 cm⁻¹ is related to the stretching vibration ν(C—O) in η²-adsorbed acetaldehyde, 1058 cm⁻¹ corresponds to the stretching vibration ν(C—O) of bidentate-adsorbed acetaldehyde. For Mg₂Al-LDO, the band at 1277 cm⁻¹ is dominant, which inhibits further condensation of acetaldehyde [32]. This is in agreement with the lower C₃₊OH selectivity (Table 1). With reaction temperature increasing, the band at 2982 cm⁻¹ becomes smaller and eventually disappears, while the band at 2936 cm⁻¹ becomes larger, indicating that more CH₂ groups are formed from long carbon chain substances by condensation [52]. However, two new bands at 1642 and 1745 cm⁻¹ appear over 290 °C. The band at 1642 cm⁻¹ is assigned to the stretching vibration ν(C=O) of acrolein and butenal [33], 1745 cm⁻¹ corresponds to the linear adsorption of η¹-adsorbed acetaldehyde [32], both bands become stronger with increasing temperature, demonstrating that rising temperatures benefit dehydrogenation and the formation of more condensation products. As the temperature increases, a similar regularity is observed for

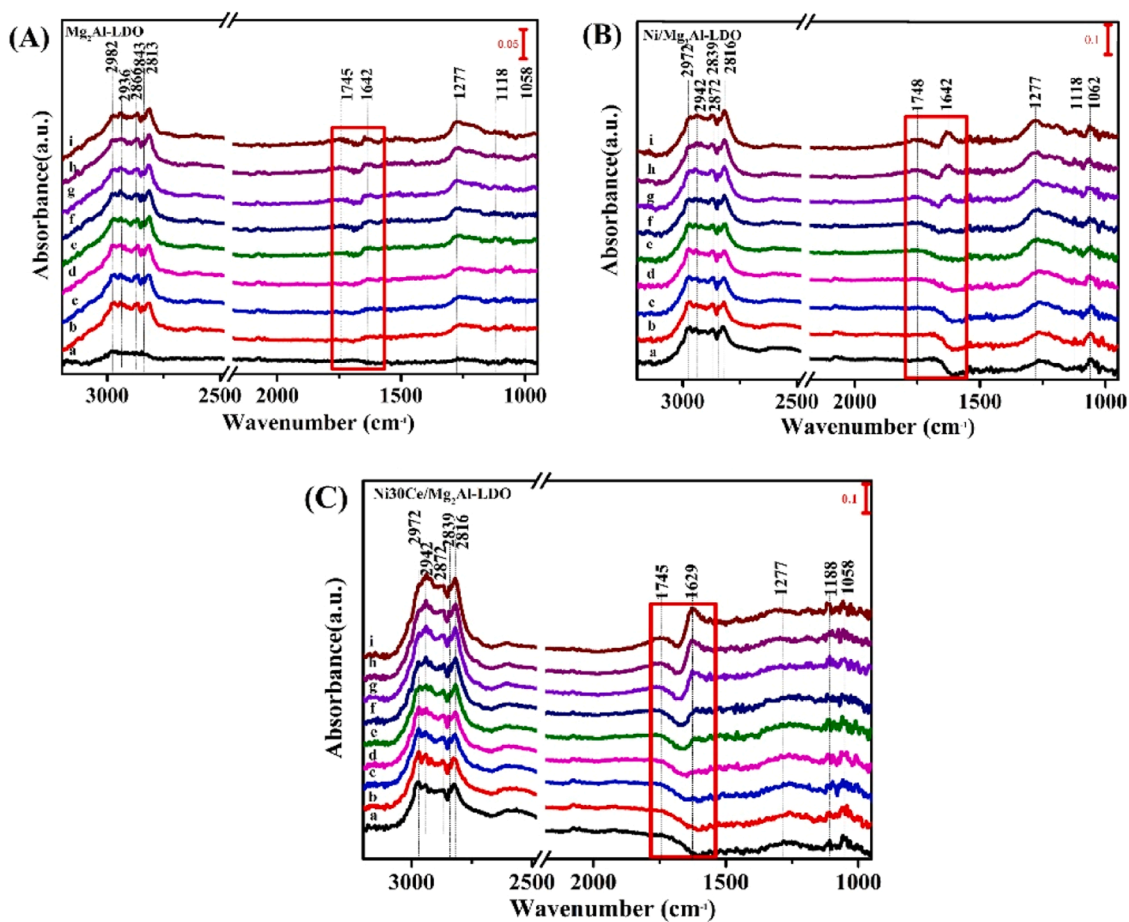
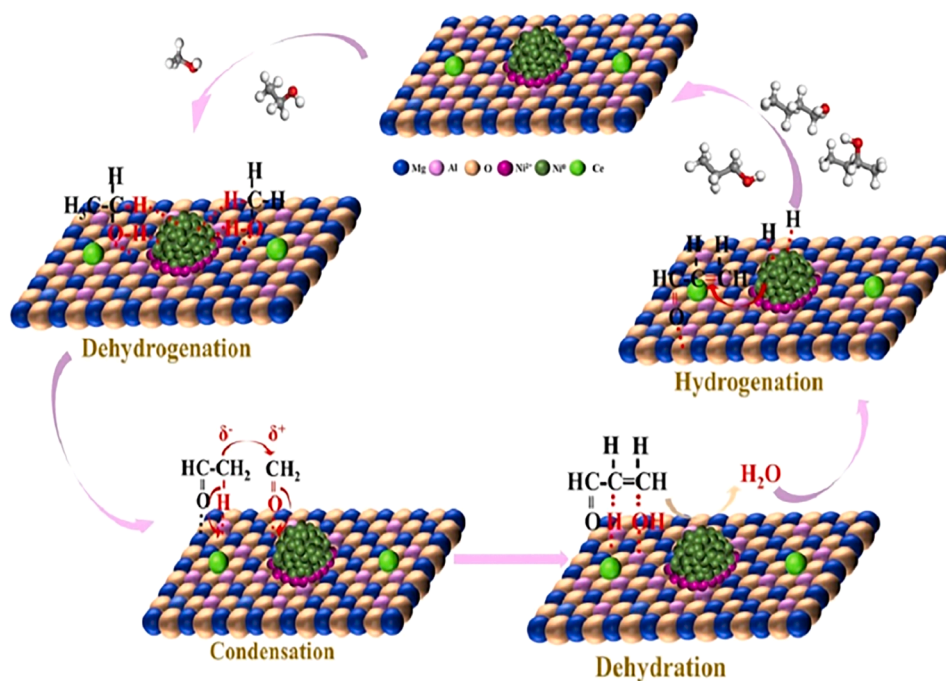


Fig. 8. In situ DRIFTS spectra of the condensation of methanol and ethanol over (A) $\text{Mg}_2\text{Al-LDO}$, (B) $\text{Ni/Mg}_2\text{Al-LDO}$ and (C) $\text{Ni}_{30}\text{Ce/Mg}_2\text{Al-LDO}$ at (a)50, (b)100, (c) 150, (d) 200, (e) 260, (f) 290, (g) 320, (h) 350 and (i) 380 °C.



Scheme 1. Proposed mechanism of the conversion of methanol and ethanol to C_{3+}OH over the $\text{Ni/Ce/Mg}_2\text{Al-LDO}$ catalyst.

Ni/Mg₂Al-LDO. Moreover, the intensities of the bands become stronger compared to those of Mg₂Al-LDO, indicating adding Ni promotes the adsorption of methanol and ethanol on the catalyst and facilitates dehydrogenation and condensation. For Ni₃₀Ce/Mg₂Al-LDO, the intensities of the bands at 2982, 2936, 2866 and 2813 cm⁻¹ are more obvious than those for Ni/Mg₂Al-LDO, and the bands at 1642 and 1745 cm⁻¹ can be clearly observed at 260 °C, indicating that the addition of Ce promotes dehydrogenation and forms more acrolein and butenal intermediates, which accounts for the higher ethanol conversion and selectivity of C₃₊OH (Table 1, entry 4). The order of strength of η^1 -adsorbed acetaldehyde at 1745 cm⁻¹ is Ni₃₀Ce/Mg₂Al-LDO > Ni/Mg₂Al-LDO > Mg₂Al-LDO, which is in accordance with the order of ethanol conversion. At the same time, the strength of C=O from acrolein and butenal at 1642 cm⁻¹ which represents the condensation ability of the catalyst, follows the order: Ni₃₀Ce/Mg₂Al-LDO > Ni/Mg₂Al-LDO > Mg₂Al-LDO, which is in agreement with the order of C₃₊OH yield (Table 1). It can be seen that, the dehydrogenation ability of Mg₂Al-LDO is insufficient to promote the subsequent condensation reaction. However, adding Ni and Ce can significantly improve the ability of acetaldehyde condensation on the surface of Ni/Mg₂Al-LDO and Ni₃₀Ce/Mg₂Al-LDO which generates more condensation intermediates (acrolein and butenal). Furthermore, the observed intermediate indicates that the coupling of methanol and ethanol over Ni₃₀Ce/Mg₂Al-LDO follows the Guerbet route.

According to the above conclusions, the mechanism of C₃₊OH synthesis from methanol and ethanol over Ni₃₀Ce/Mg₂Al-LDO is proposed (Scheme 1). First, methanol and ethanol dehydrogenate on the Ni⁰ sites to form formaldehyde and acetaldehyde, respectively. Then formaldehyde and acetaldehyde adsorb and condense on strong basic/medium-strong acid sites to form hydroxypropanal and hydroxybutyraldehyde. Finally, propanol and butanol are produced through dehydration and hydrogenation. During the reaction, Ni⁰ accelerates dehydrogenation and hydrogenation, while a moderate proportion of strong base/medium-strong acid sites promotes condensation. These results well explain the higher C₃₊OH selectivity and yield over the Ni₃₀Ce/Mg₂Al-LDO catalyst (Table 1).

4. Conclusions

In summary, methanol and ethanol were converted into C₃₊OH over Ni₃₀Ce/Mg₂Al-LDO catalysts in a fixed-bed reactor under atmospheric pressure. The Ni₃₀Ce/Mg₂Al-LDO exhibited the highest selectivity of C₃₊OH (81.0 %) with an ethanol conversion of 16.0 % at 320 °C. Characterization results reveal that the addition of Ce can not only optimize the distribution of acid-base sites, enhancing the C—C coupling ability of the catalyst, but also change the proportion of Ni⁰ in the total Ni species, which effectively facilitates the dehydrogenation and hydrogenation ability. Both the high ratio of Ni⁰ in Ni species, Ce³⁺/Ce⁴⁺ redox pair, and the moderate proportion of strong base/medium-strong acid sites are beneficial for converting methanol and ethanol into C₃₊OH. Meanwhile, the mechanism has been further elucidated by in-situ infrared spectroscopy. It shows that methanol and ethanol are firstly dehydrogenated to form formaldehyde and acetaldehyde, which then adsorb and condense on strong basic/medium-strong acid sites to form condensation intermediates (acrolein and butenal). Finally, dehydration and hydrogenation take place to produce propanol and butanol.

CRediT authorship contribution statement

Dan He: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Juan Zhou:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Kegong Fang:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.mcat.2025.115079.

Data availability

The authors do not have permission to share data.

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